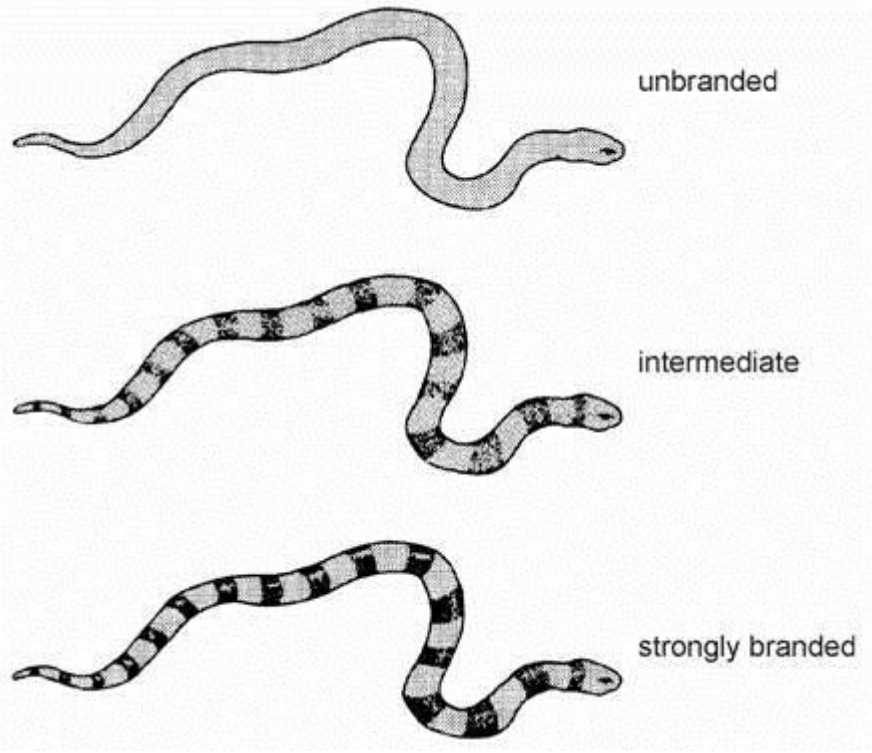
**Q1.** The water snake *Natrix sipedon* lives on small islands in Lake Erie in North America and on the nearby mainland.

The colour pattern on the body of the snakes is inherited. There are three body patterns.




A survey of the snakes on the islands gave the following results:

| body pattern | percentage of newly hatched snakes | percentage of adult snakes |
|---|---|---|
| unbanded | 8 | 37 |
| intermediate | 74 | 58 |
| strongly banded | 18 | 5 |

(a) (i) Which body pattern seems to improve the chances of survival to adulthood?

.....

1 mark

(ii) Explain how camouflage helps snakes to survive.

.....

.....

1 mark

There is much more vegetation on the mainland than on the islands.
Most of the snakes on the mainland are strongly banded.

Snakes from the mainland interbreed with snakes on the islands.

(b) (i) Suggest how the proportions of strongly banded and unbanded snakes on the islands would change over a number of years if this interbreeding stopped.

.....

.....

1 mark

(ii) Explain your answer with reference to the genes for body pattern.

.....

.....

.....

2 marks
Maximum 5 marks

## Q1.

(a) (i) unbanded

**1**

(ii) **a reference to predators or prey is required for the mark**

they are less visible to predators **or** less visible to prey

*accept 'can hide from' for 'less visible to'*
*do **not** accept 'less visible against rocks'*

**1**

(b) (i) proportion of strongly banded snakes would decrease
**or** proportion of unbanded snakes would increase

*accept 'fewer strongly banded' **or** 'more unbanded snakes'*

**1**

(ii) the unbanded snakes are more likely to survive to breed

*accept 'strongly banded snakes are less likely to survive to breed'*

**1**

genes for the unbanded pattern are inherited by higher proportion of the offspring

*accept 'gene pool on island would contain a higher proportion of unbanded genes **or** a lower proportion of banded genes'*

**1**

**[5]**